Akka vs. Databricks

Mosaic AI / Agent Bricks — a comparison for teams building agentic AI

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Databricks is a lakehouse with an agent layer attached — the data, feature, and model tier, **not a governed agentic runtime**. Mosaic AI and Agent Bricks build agents that read your Lakehouse and run on stateless serving endpoints; their governance is *data* governance (Unity Catalog), not *runtime* EU AI Act enforcement. Akka is the governed agentic runtime — durable in-memory state, a six-nines SLA, inline enforcement — and complements the lakehouse beneath it.

99.9%	99.9999%	4ms	877
DATABRICKS CONTROL-PLANE SLA	AKKA PLATFORM SLA	AKKA STATE READS	CONTROLS ENFORCED PRE-DEPLOY

DIMENSION	DATABRICKS (MOSAIC AI / AGENT BRICKS)	AKKA
What it is	A data intelligence platform (lakehouse) with an agent layer — Agent Bricks, Mosaic AI Agent Framework, Model Serving, Vector Search, Genie	A full-stack agentic systems platform — agents, orchestration, memory, streaming, APIs, and governance on one runtime
Primary role	The data, feature, and model tier agents read from	The runtime agents run on
Agent state	Stateless serving endpoints; session/long-term memory externalized to Lakebase (managed Postgres), re-read each turn	Durable sharded in-memory — 4ms reads / sub-10ms writes, replayable from the event journal
Availability SLA	99.9% control plane; Model Serving rides the Databricks SLA, no published six-nines agent-tier number	99.9999% across the entire platform, contractual, backed by indemnities
HA / DR	Customer-architected; typically active-passive; RTO < 1h, RPO < 15min (critical); Model Serving endpoints not in managed DR	Active-active; sub-1-minute RTO; zero-byte RPO — Akka owns it
Governance / EU AI Act	Data governance: Unity Catalog lineage, access control, audit; DAGF/DASF frameworks & guidance	Runtime enforcement: inline guardrails, hash-chained evidence, pre-deploy classification, sealed posture, Akka Verify
Coupling	Agents bound to the Lakehouse, Unity Catalog, and DBU-metered serving	Deploy anywhere — Akka cloud, your VPC, your Kubernetes, on-prem, sovereign cloud; portable specs
Cost model	DBU consumption metering (per-token + provisioned-throughput DBUs); scales with load	Shared compute on one runtime; up to 90% lower infrastructure for the same workload; fixed annual fee
Maturity	Agent Bricks beta Jun 2025; Supervisor Agent GA Feb 2026; evolving quickly	18 years, 100,000+ production deployments, 52 banks

1. A Data Platform With Agents, Not a Governed Agentic Runtime

Databricks is a lakehouse — Delta Lake storage, Unity Catalog governance, and Mosaic AI on top — with an agent layer attached. Agent Bricks, the Mosaic AI Agent Framework, Vector Search, and Genie build agents that read your governed data; Model Serving runs them. That is the data, feature, and model tier. It is not a runtime engineered to *run* an agentic system with guarantees. The line is structural: a lakehouse optimizes storage, retrieval, lineage, and model access; an agentic runtime optimizes durable execution, in-

memory state, failover, and inline policy enforcement on running processes. Databricks delivers the first and attaches agents to it; Akka is the second.

Capability	Databricks	Akka
Lakehouse / feature store / model serving	Yes — core strength	Not Akka's layer (complement)
Vector search / semantic retrieval	Yes — Mosaic AI Vector Search	Not Akka's layer (complement)
Native durable agents with in-memory state	Stateless endpoints; state externalized to Postgres	Built in — durable sharded in-memory
Six-nines platform availability SLA	Not published for the agent tier	99.9999%, contractual
Active-active HA/DR, sub-1-min RTO, zero RPO	Customer-architected; serving endpoints excluded from managed DR	Built in — Akka owns it
Inline runtime EU AI Act enforcement	Data governance + frameworks/guidance	Built in — aspect-woven

2. Reliability: Agent State and the SLA That Covers It

Akka publishes a **99.9999%** availability SLA across the entire platform — contractual and backed by indemnities — with sub-1-minute RTO, zero-byte RPO, and active-active HA/DR that Akka owns and operates 24/7. Databricks publishes a 99.9% control-plane SLA; Model Serving is "backed by the Databricks SLA," but no six-nines number is published for the agent-serving tier. Disaster recovery is customer-architected — the documented pattern is active-passive, with targets of RTO under 1 hour and RPO under 15 minutes for critical workloads — and Model Serving endpoints are **not replicated in Databricks managed disaster recovery**.

The deeper difference is agent state. Databricks agent runtimes are stateless by design: state "must be externalized to a durable store," so the entire session history is retrieved from a central database (Lakebase managed Postgres) at the start of every turn. Akka holds agent state in durable sharded in-memory storage — 4ms reads, sub-10ms writes, replayable from its event journal — so working memory survives failover without a per-turn database round trip.

Metric	Databricks	Akka
Availability SLA	99.9% control plane; no published agent-tier six-nines	99.9999% — entire platform
Allowed downtime / year	~8.7 hours (at 99.9%)	~31 seconds
RTO	< 1 hour (customer-architected, critical workloads)	Sub-1 minute
RPO	< 15 minutes (customer-architected)	Zero byte
HA/DR posture	Typically active-passive; serving endpoints not in managed DR	Active-active, Akka-owned
Agent state	Stateless; externalized to Postgres, re-read each turn	Durable in-memory, 4ms / sub-10ms

3. Up to 90% Cheaper to Operate

AI systems built with Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required to run the same agentic transaction volume, not list price. The drivers are actor concurrency (~10T tokens/core/year vs ~2T; ~80% less compute than Python frameworks), shared compute, and micro-checkpointing. Manulife reported up to 300% more concurrency and 30–50% faster processing after porting from Python.

Databricks bills on **DBU consumption** — pay-per-token for Foundation Model APIs plus provisioned-throughput DBUs for hosted models and serving — so the meter moves with load, and the lakehouse, Vector Search, and serving each consume separately. Akka runs all of it on one shared-compute runtime for a fixed annual fee finance can forecast.

4. Data Governance vs. Runtime Enforcement

Databricks' governance is **data governance**. Unity Catalog enforces access control, captures column-level lineage, logs activity for audit, and now governs agents, tools, and models as catalog assets with on-behalf-of access controls. Databricks also publishes responsible-AI guidance (the AI Governance Framework: 5 pillars, 43 considerations; the AI Security Framework). These govern the data and access plus best practice — not inline enforcement on the running agent.

The EU AI Act expects enforcement at the runtime: prohibited practices blocked as they occur, immutable records witnessed as decisions happen, human override of running processes, pre-deployment classification, and a retained evidence trail. Akka embeds all of this in the runtime.

5. Governance and the EU AI Act

Akka enforces governance in the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; **hash-chained immutable evidence**; HITL/HOTL control; pre-deployment classification against 186 regulations and 877 controls (288 carrying a financial penalty); multi-persona sign-offs; a sealed Governance Posture Package; and Akka Verify, which proves conformance from the running system. Unity Catalog governs the data the agent reads; Akka governs what the agent *does*.

The penalties are enforceable now

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover
Incorrect / misleading information	EUR 7.5M or 1.5% global turnover

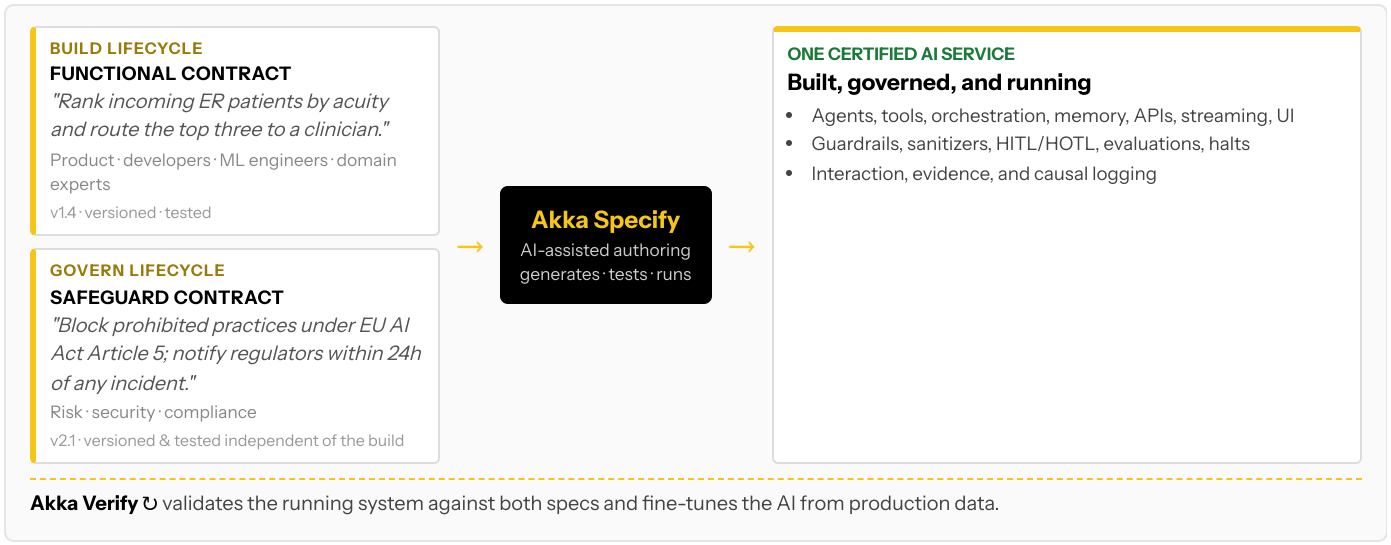
High-risk AI carries a 10-year logging-retention obligation (Art. 72), enforceable since Feb 2025 / Aug 2025.

Data governance is not runtime enforcement

Unity Catalog answers *who may read which data and where it came from.* The EU AI Act asks *was a prohibited decision blocked as it happened, was a human able to override the running process, was the system classified before it shipped, and is the evidence sealed and immutable.* Frameworks and lineage do not gate a deployment or halt a running agent; Akka enforces that inline.

6. Two Lifecycles, One Certified System

Building agents on Databricks is a developer-and-data-team workflow: define an agent in Agent Bricks, wire it to Lakehouse context and Unity Catalog permissions, deploy to Model Serving, evaluate. Governance runs as a parallel data-governance and guidance track. Akka runs two independent lifecycles on one platform via **Akka Specify**:



The build and governance lifecycles are versioned and tested independently, by different audiences — a workflow Databricks runs as data governance plus guidance, not as an enforceable safeguard contract on the running agent.

7. Real-Time Streaming at Petabyte Scale

Akka's streaming is built into the runtime — continuous, backpressured, **petabyte-scale in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second). On Databricks, real-time data movement lives in the lakehouse (streaming tables, pipelines); feeding it into the agent layer is an integration the customer assembles across serving endpoints, Vector Search sync, and Lakebase, each metered and operated separately.

8. For the Buyer: Product Maturity, Coupling, and Accountability

The Databricks lakehouse is mature and widely adopted. The question for an agentic-AI decision is the **agent product's** maturity and what standardizing on it commits you to — not Databricks' viability, which is not in question.

Buyer concern	Databricks (Mosaic AI / Agent Bricks)	Akka
Agent product maturity	Agent Bricks announced beta Jun 2025; Supervisor Agent GA Feb 2026; capabilities and APIs evolving quickly	18 years; 100,000+ production deployments; 52 banks
Coupling / lock-in	Agents bound to the Lakehouse, Unity Catalog, and DBU-metered serving; value compounds as more of the estate sits in Databricks	Deploy anywhere — Akka cloud, your VPC, your Kubernetes, on-prem, sovereign cloud; portable specs; BSL licensing
Who owns the SLA	Customer architects HA/DR; serving endpoints excluded from managed DR	Akka owns the SLA, 24/7 SRE, one platform
Certifications	SOC 2 Type II, ISO 27001, ISO 27701, PCI DSS, HIPAA, HITRUST (Azure)	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, CCPA, PIPEDA, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Risk transfer	Standard cloud terms	Availability and data-integrity guarantees backed by contractual indemnities
Budget predictability	DBU consumption metering that scales with load	Fixed annual fee finance can forecast

Akka complements your lakehouse

Akka is explicitly **not** a vector database, a semantic knowledge layer, a model-serving / inference engine, or a context graph — those are exactly what the Databricks lakehouse, Unity Catalog, Vector Search, and Model Serving provide. For those layers Databricks sits *beneath* Akka, and the two are complementary: keep your governed data and feature/model tier in Databricks, and run the governed agentic system on Akka above it. The competitive overlap is narrow and specific — Agent Bricks and the Mosaic AI Agent Framework as the agent runtime and governance layer — and that is where the runtime-versus-data-platform line is drawn.

Customers Running Agentic and Real-Time Systems on Akka

Manulife
2,000
developers across 100 projects on one governed platform

Tubi
5B tok/s
real-time hyper-personalization engine

Swiggy
71ms
order-assignment AI, ~50% latency reduction

John Deere
1,000+
tractor sensors turned into real-time insight

Verizon
750%
order-processing capacity gain; 6s → 2.4s response

Common Questions

We already run Databricks. Why add Akka?
Keep it. Databricks is a strong lakehouse, and Akka complements it — Akka is not a vector DB, model-serving, or semantic layer. The gap is the agentic runtime: durable in-memory agent state, a six-nines platform SLA, active-active failover, and inline EU AI Act enforcement. Run your governed data and models on Databricks and the governed agentic system on Akka above it.

Doesn't Databricks have agent memory now?
Databricks agents are stateless and externalize state to Lakebase (managed Postgres), re-reading session history from a central database at the start of every turn. Akka holds agent state in durable sharded in-memory storage — 4ms reads, sub-10ms writes, replayable from its event journal — so working memory survives failover without a per-turn database round trip.

Agent Bricks is "governed." Isn't that the same governance?
Agent Bricks governs agents, tools, and models as Unity Catalog assets — that is data governance: access control, lineage, and audit, with on-behalf-of permissions. The EU AI Act expects runtime enforcement: prohibited practices blocked inline, immutable records witnessed as decisions happen, human override of running processes, pre-deployment classification, and a sealed evidence trail. Akka enforces these in the runtime; Unity Catalog governs the data the agent reads.

Is Databricks cheaper because we already pay for it?
Databricks bills on DBU consumption — pay-per-token plus provisioned-throughput DBUs for serving — so the agent meter scales with load and each layer consumes separately. Akka runs the whole agentic system on one shared-compute runtime, up to 90% cheaper to operate for the same workload, on a fixed annual fee.

Sources

Agent Bricks (what / GA): databricks.com/blog/introducing-agent-bricks (beta, Jun 2025); databricks.com/blog/agent-bricks-supervisor-agent-now-ga-orchestrate-enterprise-agents (Supervisor Agent GA, Feb 10 2026); developers.databricks.com/docs/agents/overview — agents/tools/models governed via Unity Catalog + on-behalf-of access
Mosaic AI Agent Framework / Model Serving: databricks.com/product/model-serving; docs.databricks.com/.../agent-framework/deploy-agent — agents deployed to serving endpoints; "backed by the Databricks SLA," serverless HA
Agent state / memory: docs.databricks.com/.../agent-framework/stateful-agents; learn.microsoft.com/.../oltp/projects/state-management — "agent runtimes are typically stateless"; state externalized to Lakebase Postgres, session history retrieved each turn
Unity Catalog / Vector Search / Genie: databricks.com/product/unity-catalog — centralized access control, lineage, audit (data governance); learn.microsoft.com/.../vector-search/vector-search
SLA / DR: docs.databricks.com/.../reliability/best-practices — 99.9% control-plane SLA; docs.databricks.com/aws/en/admin/disaster-recovery & managed-disaster-recovery — typically active-passive, RTO < 1h / RPO < 15min (critical), "Model serving endpoints are not replicated in Databricks managed disaster recovery"
Pricing (DBU): databricks.com pricing; flexera.com/blog/finops/databricks-pricing-guide — DBU consumption; pay-per-token Foundation Model APIs + provisioned-throughput DBUs for serving
Governance / EU AI Act: databricks.com/trust/responsibleAI; databricks.com/blog/introducing-databricks-ai-governance-framework — DAGF (5 pillars, 43 considerations), DASF; committed to EU AI Act compliance
Certifications: databricks.com/legal/security-addendum; databricks.com/trust/compliance/soc — SOC 2 Type II, ISO 27001, ISO 27701, PCI DSS, HIPAA; HITRUST (Azure Databricks)
Akka facts (akka-facts.md, verified 2026-06-19): 99.9999% availability, active-active HA/DR, sub-1-min RTO, zero-byte RPO, contractual indemnities; durable in-memory 4ms / sub-10ms, replayable journal; ~10T vs ~2T tokens/core, ~80% less compute, up to 90% cheaper (akka.io/blog/go-slow-to-go-fast); Manulife up to 300% more concurrency, 30–50% faster; 186 regulations / 877 controls / 288 with financial penalty; 19 standards (trust.akka.io); 18 years, 100,000+ deployments, 52 banks; profitable, Dell Technologies Capital; Tubi 5B tok/s, Swiggy 71ms, John Deere 1,000+ sensors, Verizon 750%